ZHOUYI HE

he@ipfdd.de ◇ +49 1744212073 ◇ Personal Page ◇ Google Scholar
Leibniz Institute of Polymer Research, 01069 Dresden, Germany

ACADEMIC TRAINING

B.Sc. in Chemistry, University of Science and Technology of China *2015 - 2019*

M.Phil. in Chemistry, Hong Kong University of Science and Technology *2019 - 2022*
Thesis: Statistical Methods for Protein Function and Evolution Analysis

Ph.D. in Physics, Leibniz Institute of Polymer Research Dresden *2023 - 04/2027 expected*
Thesis: Protein Architecture as a Regulator of Biomolecular Condensation in Cells
Supervisor: Prof. Jens-Uwe Sommer and Dr. Tyler Harmon

Visiting Scholar, Prof. Eugene Shakhnovich's Group, **Harvard University** *9-12/2025*

PUBLICATIONS

Zhouyi He, *et al.*, and Tyler S. Harmon. "Born to Condense: Polysomes Drive Co-Translational Condensation of Biomolecular Condensate Proteins." PRX Life (2026).

Zhouyi He, Jens-Uwe Sommer, and Tyler S. Harmon. "The Impact of Coiled-Coil Domains on the Phase Behavior of Biomolecular Condensates." ACS Macro Letters (2025).

M. Hasan*, **Zhouyi He***, *et al.* "Dynamic Expedition of Leading Mutations in SARS-CoV-2 Spike Glycoproteins." Computational and Structural Biotechnology Journal (2024). *: Equal contribution.

Xiaolong Yang, **Zhouyi He**, and Xiao Zheng. "Unit Cell Consistency of Maximally Localized Wannier Functions." Electronic Structure (2020).

RESEARCH ACCOMPLISHMENTS

- Prof. Xiao Zheng, USTC *2018-2019*

Unit cell consistency of maximally localized wannier functions
Wannier functions describe electronic structure using localized orbitals and are usually constructed from unit cells; this project investigates consistency and reliability of maximal localized wannier functions when constructed from super cells, i.e., multiple unit cells viewed as one single cell.

- Prof. Haibin Su, HKUST *2020-2023*

Sequence-dependent target binding and cleavage kinetics of CRISPR Cas9
CRISPR Cas9 off-target effects can be viewer as error tolerance. Built a kinetic model from guide-target binding to DNA cleavage including mismatch-dependent proofreading, and used it to interpret how mutations in DNA-binding regions improve target fidelity by weakening non-specific interactions.

Evolution of CRISPR Cas systems in Streptococcus genus
CRISPR-Cas is a bacterial defense system storing "memory" for future protection. Applied statistical coupling analysis to identify co-evolving sectors in 3D protein structures, revealing an evolutionary trade-off between horizontal gene transfer benefits and antiviral defense strength.

Dynamic expedition of leading mutations in SARS-CoV-2 spike proteins
Developed *deLemus*, a novel time-resolved statistical method to quantify spike protein evolution dynamics. By combining single–amino-acid polymorphisms with mutation-matrix decomposition, identified dominant mutation patterns and emergent variants to inform spike-targeted therapeutic design.

RECENT PROJECTS

-Dr. Tyler Harmon, Prof. Jens-Uwe Sommer, IPF Dresden *2023 - now*

Protein architecture as a control parameter of phase separation

Structured domains are common in phase-separating proteins. Investigated how coiled-coil domains control phase behavior and material properties of condensates via coarse-grained simulations, providing a physical basis for how domain architectures tune phase separation ability and internal organization.

Co-translational condensation of polysome and protein

Found that phase separation can be coupled to translation via polysomes (many ribosomes on one mRNA). Developed a multiscale modeling framework combining coarse-grained simulation with reaction-diffusion kinetics to understand the co-localization of polysomes and condensates. Found that this is controlled by protein architecture, it can happen through polysomes nucleate condensates or diffuse to existing condensates, and it facilitates post-translational modifications.

Transcription factor protein condensates on DNA

Transcription factors are key regulators of gene expression and tightly related to biomolecular condensation. Aim to understand the co-condensation of transcription factors on DNA, characterizing the transition from one-layer binding saturation to condensation on DNA.

- Prof. Eugene Shakhnovich, Dr. Aniruddha Chattaraj, Harvard University *2025 - now*

The role of spacer length in multi-component condensate organization

Used coarse-grained simulations of sticker-spacer proteins to elucidate how spacer length governs internal condensate architecture and heterogeneity, aiming to understand mechanism and establish design principles for spatial patterning in synthetic and biological multi-component condensates.

CONFERENCE PRESENTATIONS

LLPS2026 Summer School and Workshop, Lund University, Sweden (2026).
German Physical Society (DPG) meeting (2024, 2025, 2026).
EMBO-EMBL Symposium "Cellular mechanisms driven by phase separation" (2026).
Emergence and Self-organization in Biological Systems, LMU Munich (2025).
19th Dresden Polymer Discussion Biomolecular Condensates and Polymer Phase Transitions (2025).
EMBO Workshop CELLULAR MATTERS: A deeper look into the complex cytoplasm (2025).

TECHNICAL SKILLS

Methods: Coarse-Grained Modeling, Monte Carlo Simulation, Molecular Dynamics, Reaction-Diffusion Modeling, Cahn-Hilliard equations, Machine Learning (basics), Bioinformatic Analysis, etc.
Programming: FORTRAN, Python, MATLAB, C, Bash, LaTeX, Git, HPC, LAMMPS, VMD, etc.
Expertise: Theoretical Biophysics, Statistical Mechanics, Polymer Physics, Phase Separation, Bioinformatics, Computational Biology, Reaction Kinetics, etc.

AWARDS & ADDITIONAL ACTIVITIES

Awards: Research Full Scholarship (HKUST), Outstanding Graduate (USTC), National Endeavor Scholarship (USTC), 3rd prize in China High School Chemistry Olympic Competition
Academic Service: PhD Representative (IPF Dresden), Teaching Assistants (Soft Condensed Matter Theory, Computational Soft Matter Physics, Physical Chemistry, Mathematics for Chemists, etc.)
Leadership: Church Youth Group Leader, Basketball Team Captain, Rural Teaching Program Leader

HOBBIES

Cycling, Basketball, Photography, Literature, Music, Movie, Hiking, etc.